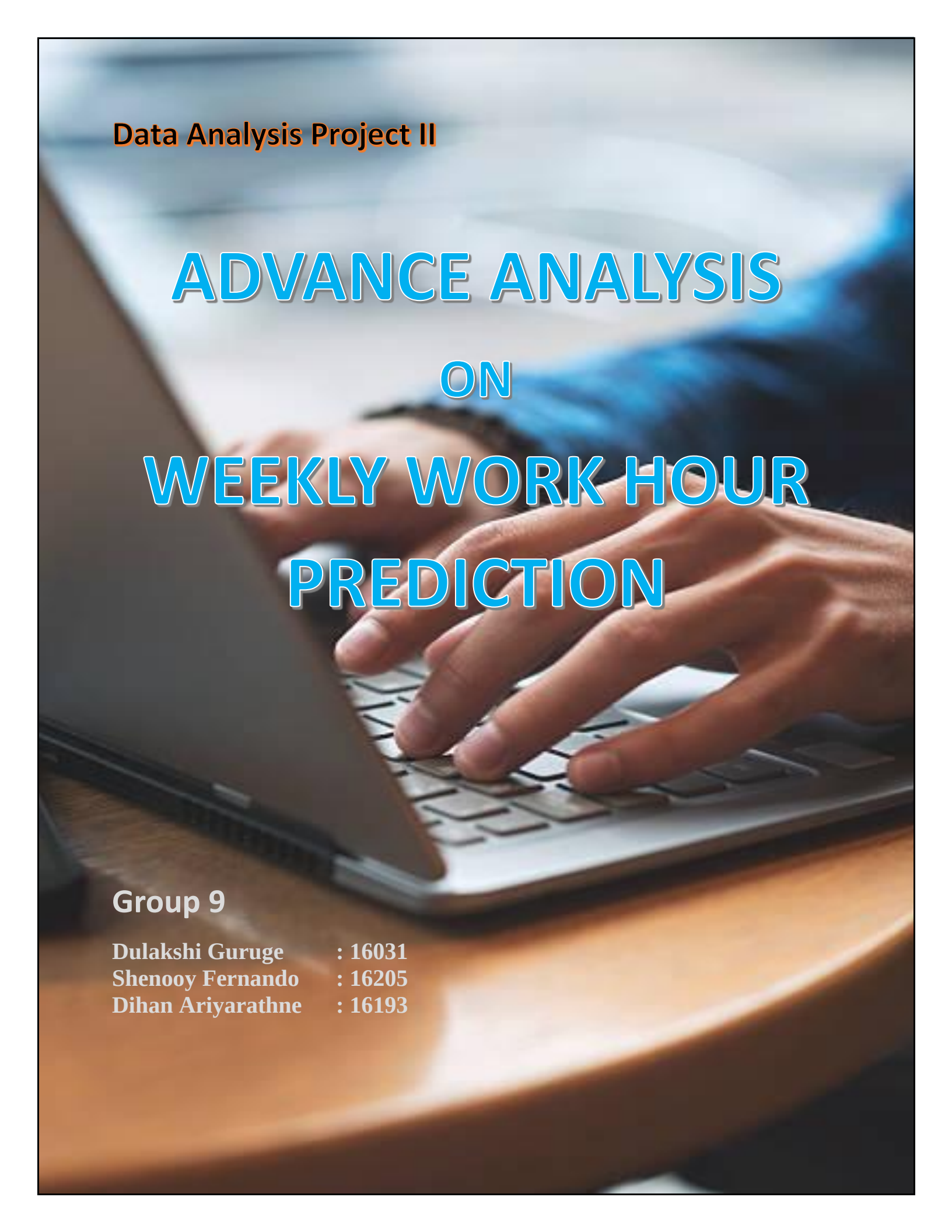




Data Analysis Project II

ADVANCE ANALYSIS ON WEEKLY WORK HOUR PREDICTION

Group 9

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Abstract

In today's dynamic work environment, understanding the factors that influence employees' weekly working hours is crucial for enhancing productivity and ensuring well-being. This report presents the findings of an exploratory data analysis conducted on the "Work Hour Prediction" dataset obtained from Kaggle. The primary objective of this analysis was to examine how weekly working hours are associated with various demographic and occupational factors. Additionally, the report provides preliminary insights into fitting a predictive model for the dataset and identifies key features that contribute to an optimal model for accurately forecasting individual working hours.

“Time is the most valuable thing a man can spend.” – Theophrastus

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1. Introduction

In today's diverse and competitive career landscape, many individuals face significant challenges in balancing their professional and personal lives. The term "work hours" refers to the time people allocate to their paid professional activities within a specific period. Technological advancements, demographic shifts, and social transformations have prompted a reevaluation of the traditional eight-hour workday. Various social, demographic, and job-related factors influence an individual's working hours. Analyzing these factors is essential for enhancing workforce productivity and addressing employees' needs in achieving work-life balance. This study aims to comprehensively understand the variables associated with an individual's working hours by conducting an exploratory data analysis on the provided dataset.

2. Description of the Question

In recent years, there has been a growing interest in understanding the factors that influence employees' weekly working hours. This interest arises from considerations related to work-life balance, productivity, overall well-being, job satisfaction, and economic implications. Understanding these factors can help organizations optimize workforce management, improve employee retention, and develop policies that promote a healthier work environment.

Hence this report aims to achieve the following:

- **Exploring factors influencing an individual's weekly working hours.**
- **Identifying overworked individuals.**

Finally, this analysis aims to develop a suitable statistical model to predict an individual's weekly working hours based on their relevant factors.

3. Description of the Dataset

The Work hour prediction data set was taken from Kaggle. It contains 18944 observations with 15 number of variables. One of them is "Hours-per-week" which is the response variable for the study. Here is the description of each variable

Variable	Description
Quantitative Variables	
Age	Age of an individual in years
Fnlwgt	Weight assigned to each individual by the census to make the dataset representative of the entire population.
Education-num	Number of years spent for education
Capital-gain	Capital gain for an individual

Capital-loss	Capital loss for an individual
Hours-per-week	Work hours per week in hours of an individual
Qualitative Variables	
Work-class	Working type of an individual
Education	Highest education qualification
Marital status	Marital Status of an individual
Occupation	Occupation of an individual
Relationship	Relationship type of an individual
Race	Type of the race of an individual
Sex	Type of the sex of an individual
Native country	Native country of an individual
Income	Income level of an individual

Table 1 - Variable Description

4. Data Pre-processing

Missing Values: No missing values were found in the dataset.

Duplicates: No duplicate records were found in the dataset.

Not meaningful values: Since we are analyzing working hours, categories like 'never worked' might not be useful. Therefore, observations in the 'not worked' category have been removed from the work class.

Data entry errors: During exploratory data analysis, we found an unusual category, 'South' in native-country, likely due to a misclassification or an error in data entry, as 'South' does not represent a specific country (It may be a country name starting with 'south'). However, it is the second most frequent category in this variable. Given its significant proportion, we retained it to preserve data structure and distribution.

4.1. Feature Engineering

➤ Recategorized Variables:

Work Class: 7 categories (without removed one) → 4 categories(private, without pay, government, self-employed)

Education: 16 categories → 4 categories (associate degree holder, degree holder, high school graduate, school level)

Marital-Status: 7 categories → 3 categories (married, separated/widowed, unmarried)

Race: 5 categories → 3 categories (white, black, other)

Native country: 41 categories → 2 categories (USA, Not USA)

- The dataset was reduced from 18944 to 18881
- Then, the dataset was split into training and test sets. The training set consisted of 15105 observations. Descriptive analysis was conducted using this training set.

And further...

- To achieve our objective of detecting overworked individuals, we further categorized the 'Hours per Week' variable into two groups: ≤ 40 and >40 .

5. Main Results of Descriptive Analysis

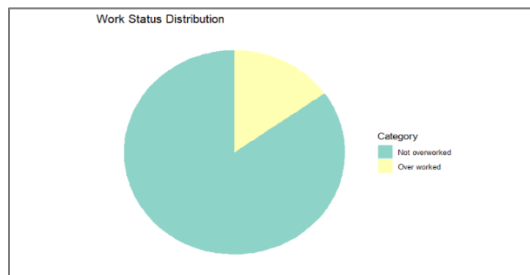


Figure 1 - Work status Distribution

Variable	GVIF	Df	GVIF ^{1/(2*Df)}
Age	1.486221	1	1.219107
Class	1.372480	3	1.054187
Fnlwgt	1.053859	1	1.026577
Education	2.499014	3	1.164225
Education number	2.250960	1	1.500320
Marital status	6.398835	2	1.590469
Occupation	2.498132	13	1.033841
Relationship	9.276626	5	1.249508
Race	1.141227	1	1.068282
Sex	1.769764	1	1.330325
Capital gain	1.085197	1	1.041729
Capital loss	1.013163	1	1.006560
Native country	1.154911	1	1.074668
Race	1.546118	1	1.244234

Table 2 - GVIF Values

- We observed multicollinearity between 'Education-num' and 'Education,' as well as between 'Relationship' and 'Marital Status.' To improve further modeling, we can remove one of each pair.

➤ It was observed that the distribution of working hours per week is approximately symmetric.

➤ Some predictor variables were observed to be skewed with some outliers.

Multicollinearity

The GVIF analysis shows strong multicollinearity between relationship status (GVIF=9.28) and marital status (GVIF=6.40) variables, both exceeding the critical threshold of 5. This indicates these variables contain overlapping information, suggesting one should be removed from the model to improve its reliability

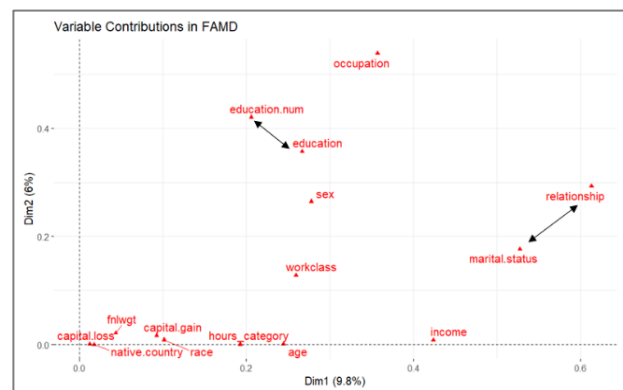


Figure 2 - Variable Contribution in FAMD

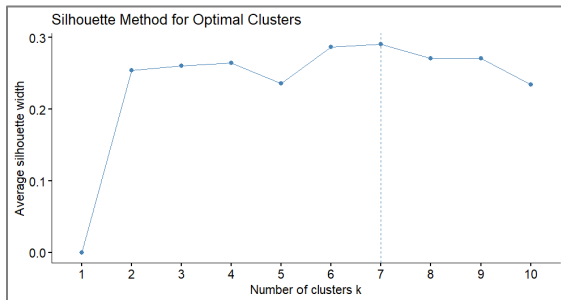


Figure 3 - Silhouette Method for Optimal Clusters



Figure 4 - Cluster Plot

- Additionally, our analysis indicates that no distinct clusters are present in the data.

Outlier Analysis

Outlier detection is vital for preventing extreme values from skewing analysis. In this study, we used two methods: Mahalanobis distance with FAMD (Factorial Analysis of Mixed Data) and Isolation Forest. The Mahalanobis distance, calculated using FAMD to account for both categorical and continuous variables, flagged observations that significantly deviated from the expected multivariate distribution as anomalies.

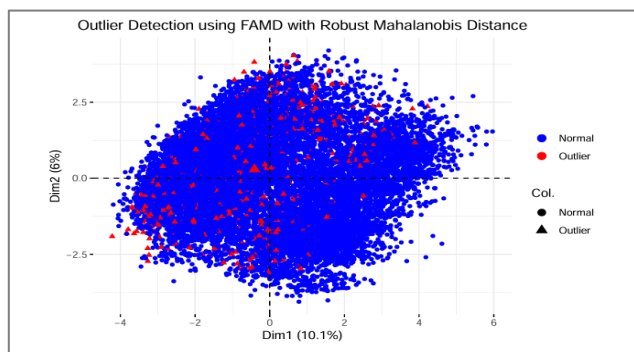


Figure 5 - Outlier Detection (Robust Mahalanobis Distance)

By this graph we identified outliers as shown in the two principal FAMD dimensions. These outliers are visualized as distinct points that fall outside the general cluster of the normal values.

We identified 756 anomalies but kept them in the dataset to avoid losing valuable information. Removing them could distort the

- By this plot we can see the outliers and the normal data points, where the outliers are flagged using Mahalanobis distance. This visualization allows for a clear comparison of the extreme values (outliers) against the regular observations.

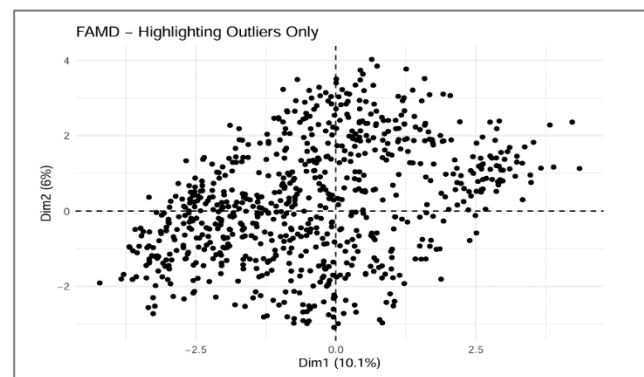


Figure 6 - FAMD - Highlighting Outliers Only

analysis, as they may represent meaningful variations. Keeping these outliers ensures a more complete dataset and helps build robust models that account for both typical and extreme cases

6. Main Results of Advance Analysis

Logistic Regression

Choice of Logistic Regression

- Binary Response Variable: The response variable is binary (≤ 40 or > 40 hours per week), making logistic regression a suitable classification technique.
- Advantages: Logistic regression is interpretable, provides odds ratios for explanatory variables, and allows assessment of the influence of various demographic and occupational factors on long working hours.
- To address multicollinearity, we removed the education number and relationship variables before applying logistic regression.

Model Performance

Matrix	Train	Over Worked	Not Over Worked	Test	Over Worked	Not Over Worked
Accuracy	0.6899	-	-	0.6843	-	-
AUC-ROC	0.7544	-	-	0.7415	-	-
Precision	0.5167	0.5167	0.8251	0.5078	0.5078	0.8231
Recall	0.6975	0.6975	0.6863	0.6929	0.6929	0.6802
F1-score	0.5936	0.5936	0.6863	0.5861	0.5861	0.7449

Table 3 - Model Performance in Logistic Regression

The evaluation metrics show consistent performance across training and test sets, with accuracy around 69% (68.99% train, 68.43% test) and AUC-ROC scores of approximately 0.75 (0.7544 train, 0.7415 test). The model demonstrates better precision in identifying non-overworked cases

(82.51% train, 82.31% test) compared to overworked cases (51.67% train, 50.78% test). Recall values remain relatively stable across both classes, hovering around 69%. The F1-scores further reflect this pattern, with stronger performance for non-overworked cases (0.7449 test) versus overworked cases (0.5861 test), suggesting room for improvement in identifying overworked individuals.

Feature Selection and Insights

After feature selection, the model achieved 73.52% training accuracy and 72.72% test accuracy. Precision improved (63.48% training, 61.19% testing), but recall remained moderate (43.45% training, 42.20% testing), leading to F1-scores of 0.5159 (training) and 0.4995 (testing). The

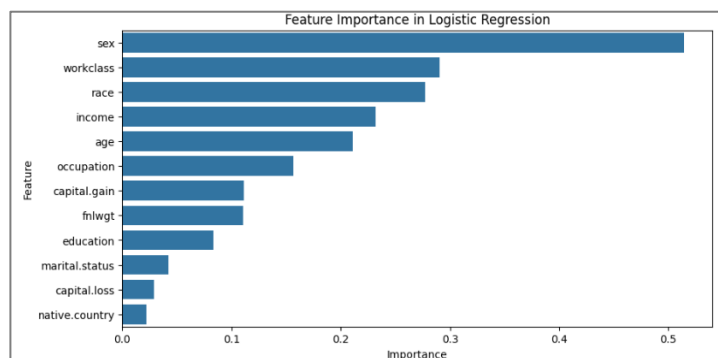


Figure 7 - Feature Importance in Logistic Regression

AUC-ROC scores (0.7543 training, 0.7417 testing) indicate reasonable discrimination, though the low recall suggests a need for resampling or threshold adjustments to better identify overworked individuals.

Conclusion

- Challenges: The model's low recall and class imbalance indicate the need for improvement.
- Future Steps:
 1. Explore resampling techniques (e.g., oversampling, undersampling) to address class imbalance.
 2. Consider alternative models, such as XGBoost, which may better handle imbalanced data.

Random Forest

Random Forest is an ensemble learning algorithm that builds multiple decision trees and combines their predictions to improve accuracy and reduce overfitting. It works by randomly selecting subsets of data and features for training each tree, ensuring diversity in predictions.

Reasoning for Model Selection

- Multicollinearity is present in the dataset, Random Forest models can battle multicollinearity well.
- Instances differing from considerably linear relationships were identified, Random Forest models can capture nonlinear and non-monotonous relationships well.

Classes	Precision	Recall	F1 Score	Support
Low	0.76	0.69	0.72	2295
High	0.58	0.67	0.62	1481

Metric	Train Set	Test Set
Accuracy	0.7276	0.6811
AUC-ROC	0.8042	0.7463

Table 4 - Model Performance in Random Forest

The random forest model was trained on resampled training data, (SMOTENC was used to oversample the minority class) and 'class weight' attribute of the model was set to 'balanced' to address the category imbalance issue. A randomized search was performed using 10 fold cross validation for the hyperparameter tuning and later was adjusted to prevent overfitting.

Feature Selection and Insights

Analysis of the Random Forest model's feature importance reveals that demographic and social factors are the strongest predictors, with sex (0.15) and relationship status (0.14) having the highest importance scores. Educational and employment-related features show moderate importance, while financial variables like capital gains and losses have minimal impact on the model's predictions.

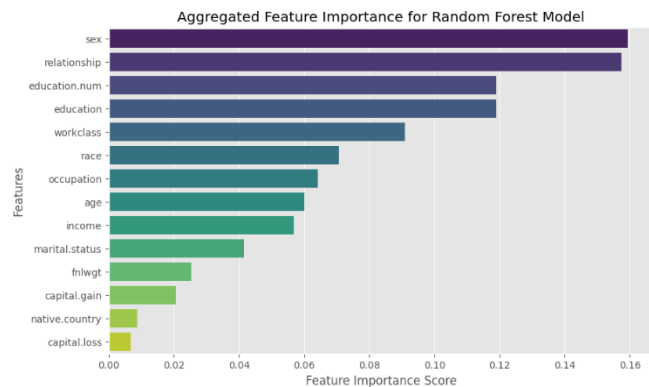


Figure 8 - Feature Importance in Random Forest

XG Boost

Extreme Gradient Boosting (XGBoost) is a powerful ensemble learning algorithm that builds multiple weak learners (decision trees) to improve classification performance. It is particularly effective in handling large datasets with imbalanced classes and multicollinearity.

Justification for Using XGBoost

- Chosen for its effectiveness with large, imbalanced datasets and ability to handle multicollinearity
- Applied to our binary classification problem (overworked vs not overworked) with 18,000 observations

Model Performance

Metric	Train	Over Worked	Not Overworked	Test	Over Worked	Not Overworked
Accuracy	0.7507	-	-	0.696	-	-
AUC-ROC	0.8263	-	-	0.7514	-	-
Precision	0.7399	0.7623	0.7399	0.7939	0.5374	0.7939
Recall	0.7686	0.7330	0.7686	0.7353	0.6170	0.7353
F1-score	0.7508	0.7473	0.7540	0.7634	0.5744	0.7634

Table 5 - Model Performance in XG-Boost

The Random Forest model achieved a training accuracy of 75.07% and test accuracy of 69.6%. SMOTE was applied to address class imbalance in the dataset. The model demonstrated strong performance in identifying non-overworked cases with 90.34% precision, though it showed lower precision (40.74%) for overworked cases, indicating room for improvement in detecting overwork conditions.

Feature Selection and Insights

The Random Forest model identified employment status, education level, and demographics as key predictors. While feature reduction was attempted based on importance scores, the full feature set was ultimately retained as removal did not improve accuracy, suggesting complex interactions between variables in the dataset

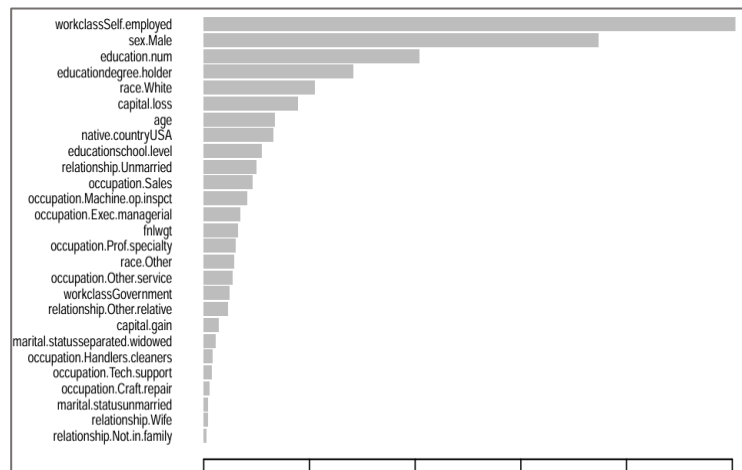


Figure 9 - Feature Importance in XG-Boost

Issues Encountered and Proposed Solutions

1. Inconsistent Data in the Native Country Variable:
 - We found "South Entry" in the *native country* variable and recoded it to "Non-USA" for consistency.
2. Multicollinearity in Predictor Variables:
 - To address multicollinearity, we removed highly correlated variables before applying logistic regression.
 - We also used robust machine learning models like XGBoost and Random Forest, which handle multicollinearity effectively.
3. Class Imbalance in Work Status:
 - The dataset was imbalanced with respect to work status.
 - To balance the classes, we applied the Synthetic Minority Over-sampling Technique (SMOTE) to generate synthetic data.
4. Outliers in the Dataset:
 - We identified potential outliers using Mahalanobis Distance.
 - However, we did not remove them, as there was no strong evidence to justify their exclusion.
5. Computational Challenges in Hyperparameter Tuning:
 - Hyperparameter tuning was time-consuming.
 - To improve efficiency, we used Random Search instead of an exhaustive grid search.

Discussion and conclusions

Model performance comparison

Model	Accuracy		F1 Score			
	Train	Test	Train		Test	
			Not Overworked	Overworked	Not Overworked	Overworked
Logistic	0.6899	0.6843	0.7493	0.5936	0.7449	0.5861
Random Forest	0.7276	0.6811	0.73	0.73	0.62	0.72
XG Boost	0.7507	0.696	0.7540	0.7473	0.7634	0.5744

Table 6 - Model Performance comparison

Our evaluation compared three machine learning models: Logistic Regression, Random Forest, and XG Boost, with a specific focus on identifying overworked individuals. While XG Boost showed the highest overall test accuracy (0.696), Random Forest demonstrated superior performance in detecting overworked cases, which is our primary objective.

Key findings:

- Random Forest achieved the highest F1 score (0.72) for overworked cases in the test set, significantly outperforming both XG Boost (0.5744) and Logistic Regression (0.5861)
- Though Random Forest's overall test accuracy (0.6811) is slightly lower than XG Boost's (0.696), its stronger performance in identifying overworked individuals makes it more suitable for our specific use case
- Random Forest maintained consistent F1 scores (0.73) across both classes in the training set, suggesting stable learning of patterns
- Our Random Forest model evaluation showed mixed results. The KS test (0.3646, $p=0.0000$) confirms strong discriminatory power, effectively distinguishing between classes. However, the Hosmer-Lemeshow test (1656.8772, $p=0.0000$) indicates poor calibration. This means while our model ranks observations effectively, its probability estimates aren't accurate. For applications requiring precise probabilities, post-processing techniques like Platt scaling could improve calibration.

Given our primary objective of identifying overworked individuals, we recommend implementing the Random Forest model. While it may have slightly lower overall accuracy, its superior performance in detecting overworked cases ($F1 = 0.72$) makes it the most appropriate

choice for this specific business problem where missing an overworked individual could have more significant consequences than a false positive.

7. Appendix including codes and technical details.

https://drive.google.com/drive/folders/1fU4KNv72arSnEwCc5mFQ3Bbt_ZaxL6YE

8. References

<https://youtu.be/STKbnti1pwY?si=E185eA5Y8gmnsVp>

https://youtu.be/qjeUhuVkbHY?si=dgHTIEVFp_NqF37S

<https://youtu.be/IXMG3ofm2KI?si=C7FIRcOSXo2r4BzK>

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